

Soft Max Classifier

The Slack Variable

As you learnt, the Support Vector Classifier allows certain points to be **deliberately misclassified**, but the Maximal Margin Classifier strictly separates the data into two halves (recall the spam-ham example).

Remember how you formulated the Maximal Margin Classifier equation in our earlier discussions (i.e. $(\sum_i (w_i \cdot y_i)) \geq M$). Similar to that, let's learn how to formulate an equation for the Support Vector Classifier.

Note : At 3:20, the subscript of w should be i instead of j .

SUPPORT VECTOR CLASSIFIER

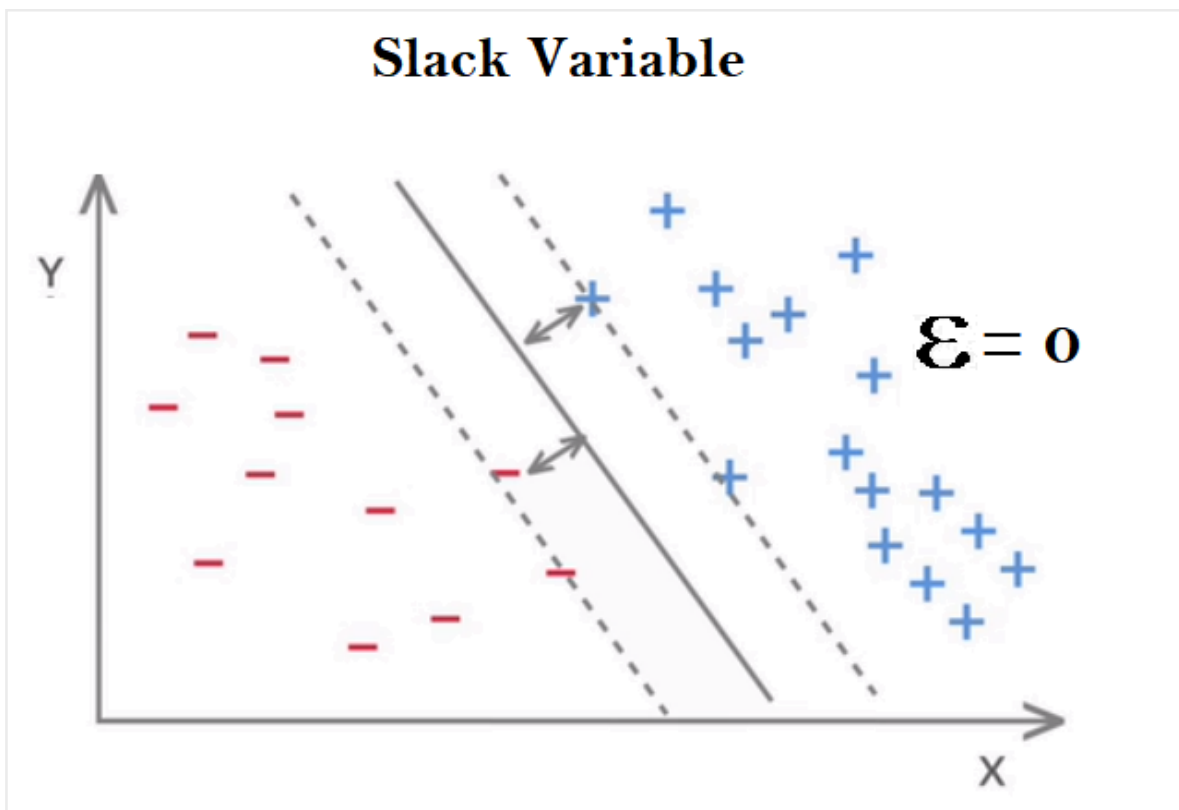
UpGrad

	1	j	..d	
	X	Y	..d	Label (l)
1	3	4	...	1
2	8	9	...	0
i	1	8	...	0
...	...	...	...	...
n	4	1	...	1

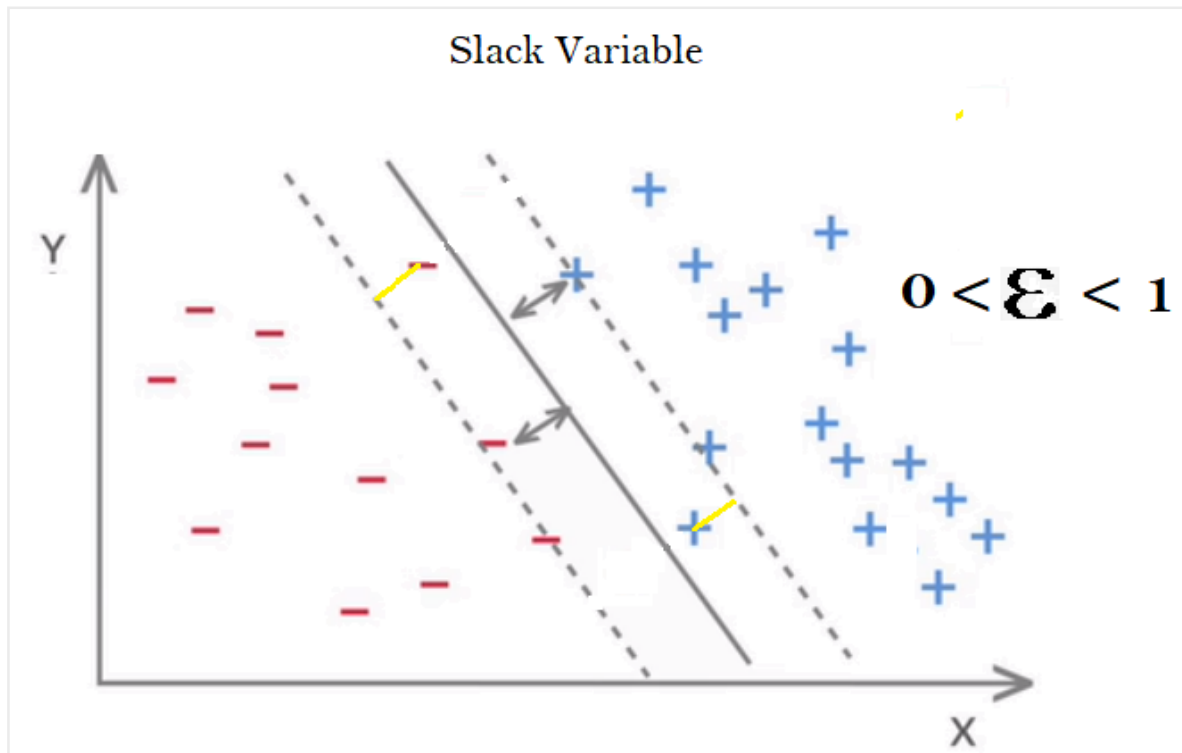
$$\sum_{i=0}^d w_j^2 = 1$$
$$i, \quad l_i \times (\vec{w} \cdot \vec{y}_i) \geq M$$
$$l_i \times (\vec{w} \cdot \vec{y}_i) \geq M (1 - \epsilon_i)$$
$$\sum_{i=0}^n \epsilon_i < c$$


We looked at the concept of the **slack variable(ϵ)**. A slack variable is used to control **misclassifications**. It tells you where an observation is located relative to the margin and hyperplane.

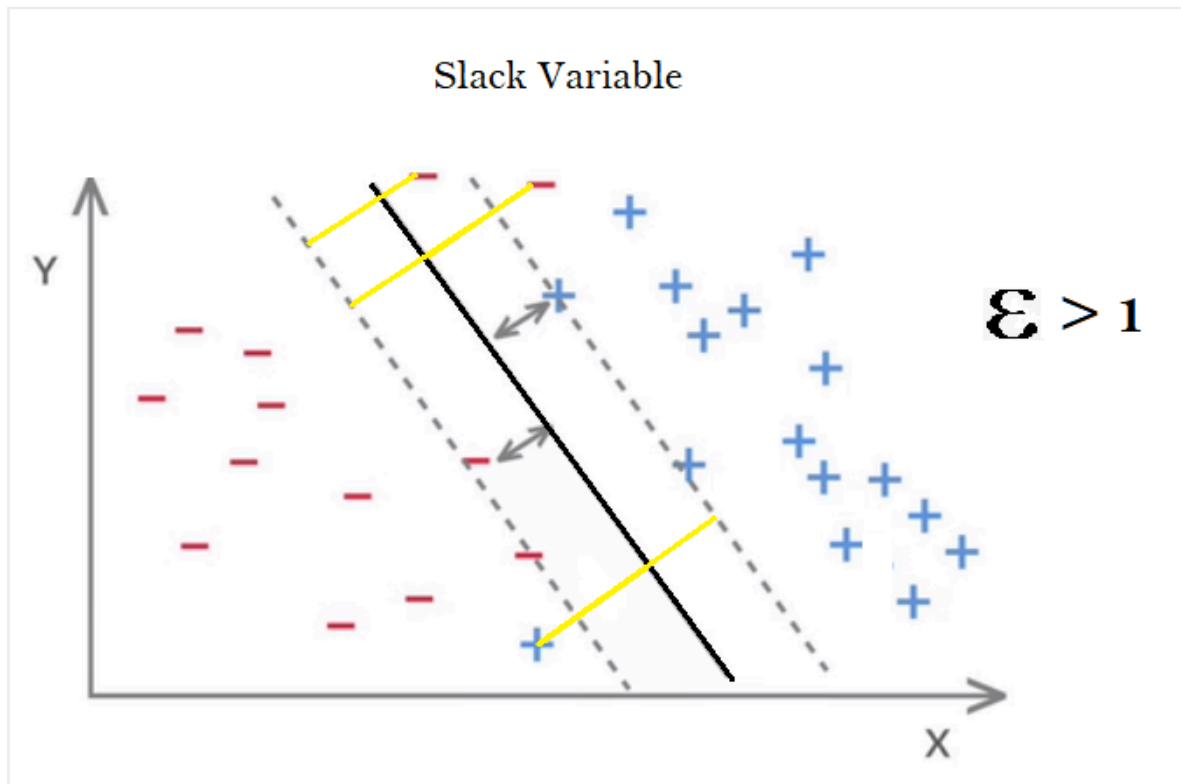
For points which are at a distance of more than M , i.e. at a safe distance from the hyperplane, the value of the slack variable is 0.



On the other hand, if a data point is correctly classified but falls inside the margin (or violates the margin), then the value of its slack ϵ is between 0 and 1. In the plot given below, the blue and the red points, which fall inside the margin, are examples of such points.



Finally, if a data point is incorrectly classified (i.e. it violates the hyperplane), the value of epsilon (ϵ) > 1 . The incorrectly classified blue and red points are such examples.



Inferences:

So you can see that

- **Each data point has a slack value** associated to it, according to where the point is located.
- The value of slack lies **between 0 and +infinity**.
- **Lower values of slack are better** than higher values (slack = 0 implies a correct classification, but slack > 1 implies an incorrect classification, whereas slack within 0 and 1 classifies correctly but violates the margin).

Once you understand the notion of the slack variable, you can easily compare any two Support Vector Classifiers. You can measure the summation of all the epsilons(ϵ) of both the hyperplanes and choose the best one that gives you the least sum of epsilons(ϵ). The summation of all the epsilons of each data point is

denoted by cost or 'C', i.e.
 $\sum \epsilon_i \leq C$.

Cost C

When **C is large**, the slack variables can be large, i.e. you allow a larger number of data points to be misclassified or to violate the margin. So you get a hyperplane where the margin is wide and misclassifications are allowed. In this case, the model is **flexible, more generalisable, and less likely to overfit**. In other words, it has a **high bias**.

On the other hand, when C is small, you force the individual slack variables to be small, i.e. you do not allow many data points to fall on the wrong side of the margin or the hyperplane. So the margin is narrow and there are few misclassifications. In this case, the model is **less flexible, less generalisable, and more likely to overfit**. In other words, it has a **high variance**.

Mathematical formulation for the Maximal Margin Classifier can be expressed as

$$l_i X(W \cdot Y_i) \geq M$$

where

- l_i represents the label of the i th observation (such as spam(+1) and ham(-1));
- **W** represents the vector of the coefficients (or weights) of each attribute (for example, if you have 3 attributes, $W = [w_0, w_1, w_2, w_3]$).
- Y_i represents the vector of the attribute values for

the i th row, e.g. $Y = [1, y_1, y_2, y_3]$ for 3 attributes.

Thus, the dot product $W.Y_i$ is simply the value of the expression obtained by putting the i th data point in the hyperplane equation, i.e. $W.Y_i = w_0 + w_1 y_1 + w_2 y_2 + w_3 y_3$.

Thus, $W.Y_i$ is lesser than, equal to or greater than 0, depending on the location of the **i th data point** with respect to the hyperplane. Also, note that the value of $W.Y_i$ gives you the distance of the **i th data point from the hyperplane**.

- **M represents the margin, i.e. the distance of the closest data point from the hyperplane.**

If you impose the condition ($\forall i, W.Y_i \geq M$) on the model, then you are implying that you want each point to be at least **a distance M away from the hyperplane**. But unfortunately, few real datasets will be so easily, perfectly separable. Thus, to relax the constraint, you include a '**slack variable**' ϵ_i for each data point i .

Thus, you modify the formulation to ($\forall i, W.Y_i \geq M(1 - \epsilon_i)$),

Where the **slack variable** (ϵ_i) takes a value between **0 to infinity**.

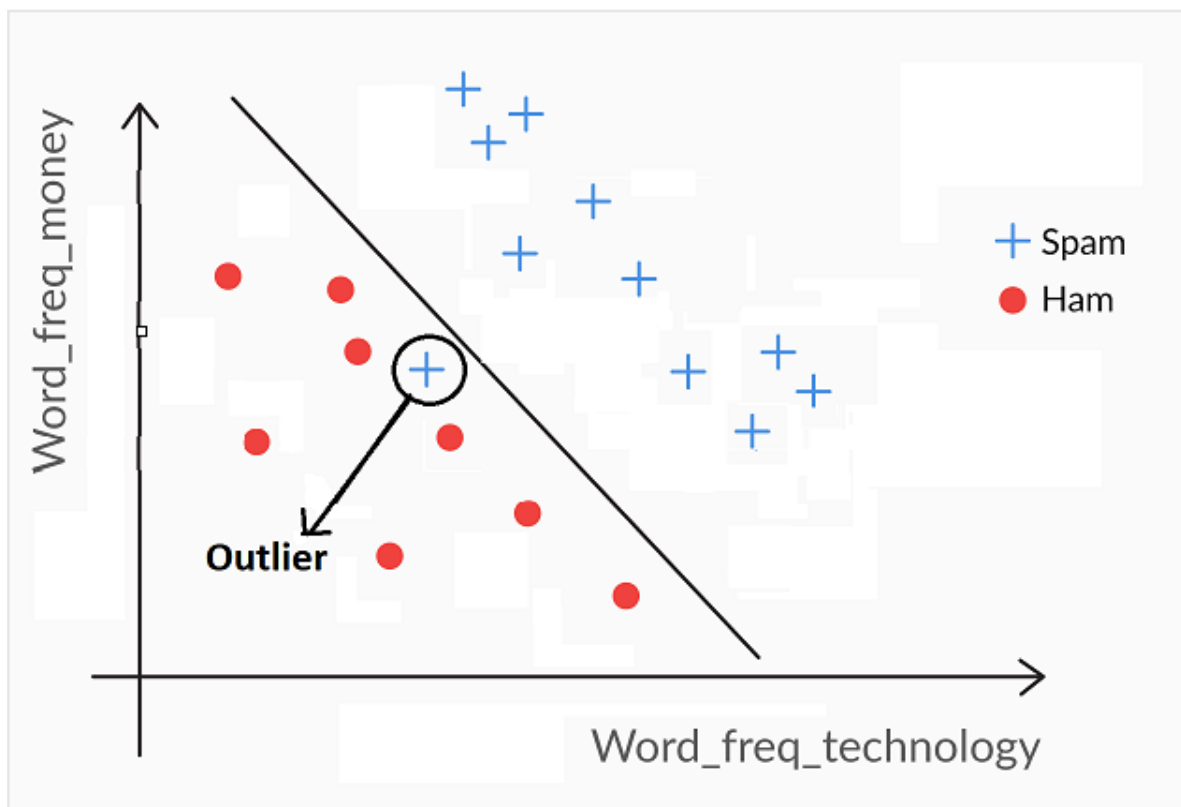
Depending on the value of ϵ_i , the i th data point can now take any position - it can fall on the **correct side of the margin** (and a safe distance away), or **inside the margin** (but still correctly classified), or **even stray on the wrong side of the hyperplane** itself.

So to summarize, misclassification can be controlled by the value of cost or C . If **C is large**, the slack variables (epsilons(ϵ)) can be large, i.e. you allow a larger number of data points to be misclassified or violate the margin; and if **C is small**, you force the individual slack variables to be small, i.e. you do not allow many data points to fall on the wrong side of the margin or the hyperplane.

So this is how the cost ' C ' influences your decision of a best-fit hyperplane. In the next lecture, you will apply the concepts you have learnt to solve a real-life problem - **classifying an email as spam or ham**.

Summary

In this session, you learnt that **Maximal Margin Classifier** has certain limitations. For instance, it will not find a separator if the classes are not linearly separable (as shown below).



The **Soft Margin Classifier** overcomes the drawbacks of the Maximal Margin Classifier by allowing certain points to be misclassified. You control the amount of misclassifications using the cost of misclassification 'C', where C is the maximum value of the summation of the slack variable ϵ , i.e.

$$\sum \epsilon_i \leq C.$$

If **C is high**, a higher number of points are allowed to be misclassified or violate the margin. In this case, the model is **flexible, more generalisable, and less likely to overfit**. In other words, it has a **high bias**.

On the other hand, if **C is low**, a lesser number of points are allowed to be misclassified or violate the margin. In this case, the model is **less flexible, less generalisable, and more likely to overfit**. In other words, it has a **high variance**.

So, C represents the 'liberty of misclassification' that you provide to the model.

Note that the C defined above and the parameter C used in the **SVC() function in python** are the inverse of each other. In `SVC()`, C represents the 'penalty for misclassification'.